

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	28.09.2026
Team ID	
Project Name	LegalEase
Maximum Marks	4 Marks

Technical Architecture

The **LegalEase** application follows a **3-tier architecture** consisting of the Presentation Layer, Application/Business Logic Layer, and Data Layer. The system supports customers, legal professionals, customer care executives, and administrators.

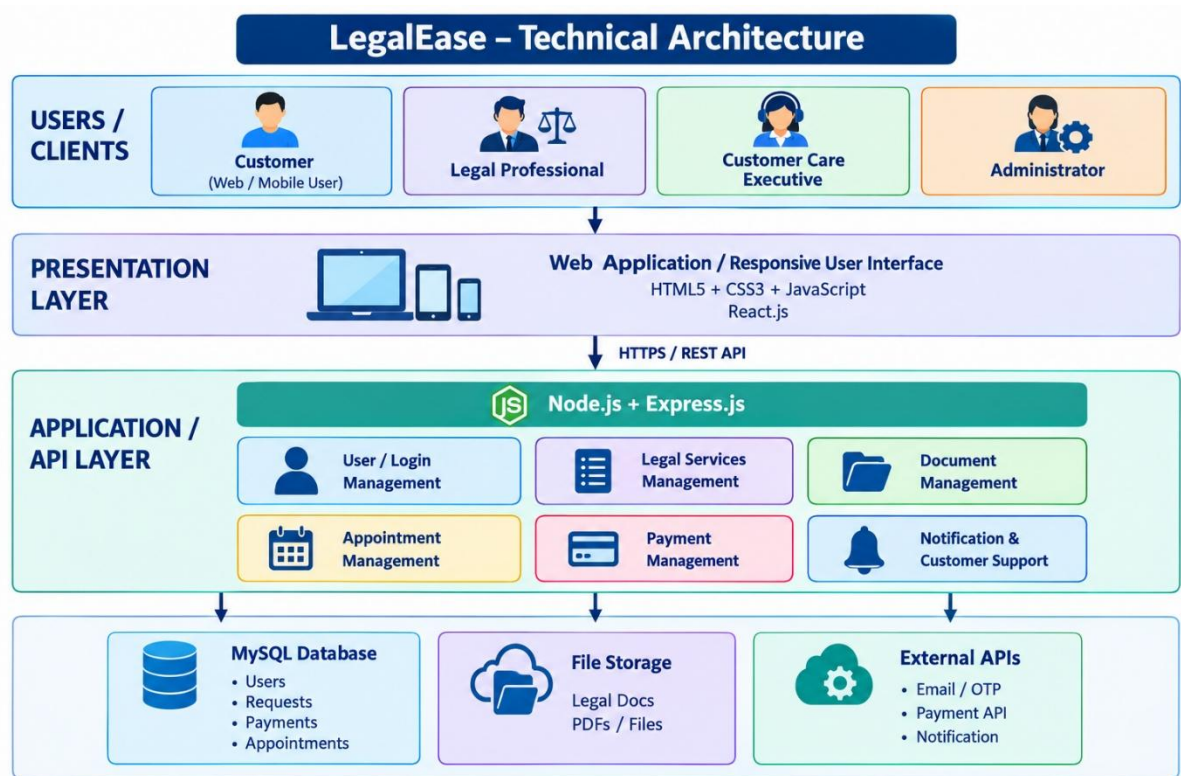


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interface through which customers, legal professionals, customer care executives and administrators interact with LegalEase.	HTML5, CSS3, JavaScript, React.js
2	Application Logic-1	Handles user registration, login, authentication, legal requests, appointments and business logic.	Node.js, Express.js

3	Application Logic-2	Handles document upload, document validation and secure document management.	Node.js / Express.js, Multer
4	Application Logic-3	Handles notifications, email confirmation, OTP and customer communication.	Node.js, Nodemailer, OTP Service
5	Database	Stores structured application data such as users, legal requests, appointments, payments and service information.	MySQL
6	Cloud Database	Cloud-based database deployment for scalable and remote access to LegalEase data.	MySQL / Amazon RDS
7	File Storage	Stores uploaded legal documents, PDFs and other files securely.	Amazon S3 / Cloud Storage
8	External API-1	Used for online payment processing for legal services.	Razorpay / Stripe API
9	External API-2	Used for sending email and OTP notifications to registered users.	Email / SMS API
10	Machine Learning Model	Optional future component for legal document classification, document categorization and intelligent search.	Python, NLP / Machine Learning
11	Infrastructure (Server / Cloud)	Hosts and deploys the LegalEase web application and backend services.	AWS / Cloud Server, Docker

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source technologies are used to develop the user interface, backend and database connectivity.	React.js, Node.js, Express.js, MySQL
2	Security Implementations	Secure authentication, password hashing, HTTPS communication, role-based access and input validation are implemented to protect user and legal information.	JWT, bcrypt, HTTPS/SSL, RBAC, OWASP practices
3	Scalable Architecture	LegalEase follows a 3-tier architecture so that the presentation, application and database layers can be scaled independently as the number of users increases.	React.js, Node.js, MySQL, Docker/AWS
4	Availability	Cloud deployment, database backup and monitoring mechanisms can be used to keep	AWS, Load Balancer, Database Backup

		the application available with minimum downtime.	
5	Performance	API optimization, database indexing, caching and optimized document handling are used to provide faster response times and support multiple simultaneous users.	Node.js, MySQL Indexing, Redis Cache, CDN